Axial Peak Position of a Chirp- and Quartic-Phase Pearcey Field with a Thin Lens

Auto BeamScientist Project

Abstract

We compare the axial location of the brightest sampled transverse pixel for an apodized Pearcey field propagated with an ideal thin lens and in free space. The registered input is modified, in order, by a quadratic chirp of +0.5 and a radial quartic phase of +0.2. Both propagation arms use the same transformed input and are evaluated over $z = 0.04$–0.08 m. In the primary calculation, the sampled peak positions are 0.064 m with the lens and 0.040 m in free space, a lens-minus-free-space difference of +0.024 m. The chirp-only ablation gives 0.060 m and 0.040 m, respectively, or +0.020 m. The supplied numerical confirmation checks pass for these two conditions. However, both free-space maxima occur at the lower boundary of the sampled interval. Under the registered interpretation rule, these boundary-selected peaks do not establish resolved physical focal positions, so the physical comparison is inconclusive. The similar chirp-only result also does not establish an incremental quartic contribution. The calculations do not isolate an operator interaction or a focusing mechanism.

1 Introduction

The axial location of a beam's brightest transverse sample describes where a sampled peak occurs; it is not interchangeable with peak intensity, collected energy, or a measure of concentration. A lens changes a field's phase and therefore its subsequent diffraction. Imposed phase structure can also alter the axial position of a sampled intensity maximum. Interpreting a shift consequently depends on the input field, the axial sampling, and whether the selected maximum lies inside the observation interval.

Pearcey beams have been studied in connection with autofocusing and self-healing [4]. Other work addresses optical catastrophe fields, including swallowtail and butterfly beams [6]. Finite-energy Airy beams and their generation and control provide a broader context for structured-beam propagation [3, 5]. The supplied bibliography identifies these topics, but does not provide full-text evidence for detailed comparisons of their field definitions, optical conditions, or observables with the present calculation.

This work addresses a condition-specific question: for an apodized Pearcey source after a quadratic chirp followed by a radial quartic phase, is the brightest sampled transverse peak located farther along the axis in an ideal 60 mm thin-lens arm than in free space? The endpoint is the difference between the two sampled peak locations over a fixed axial interval. It does not assess concentration or establish that either phase operation causes a distinct effect. A chirp-only case is an ablation, but the available conditions do not form a factorial interaction test.

2 Related Work

The Pearcey-beam literature includes work on autofocusing and self-healing [4], while the swallowtail and butterfly literature treats other optical catastrophe fields [6]. The supplied citation metadata establish these topics, but do not support a detailed comparison of the prior studies' input definitions, propagation conditions, or peak-selection procedures. We therefore make no claim of novelty relative to them.

The observable used here is the axial coordinate of the brightest sampled transverse pixel, compared between two propagation arms for the same modified input. It differs from a finite-core energy fraction, which integrates power over a selected transverse region. The supplied reference on Gaussian beams and resonators provides context for quadratic phase and Gaussian propagation [1]; the present input is instead a Pearcey field with additional phase operations. The supplied reference on Laguerre–Gaussian modes concerns orbital angular momentum and mode transformations [2], rather than this focal-position comparison.

3 Theoretical Model

3.1 Input field and phase operations

Let $V(x, y)$ denote the sampled apodized Pearcey source. The Pearcey field construction is

$$V(x,y) = C_4\Big(\frac{x}{s}, \frac{y}{s}\Big) \exp\Big(-\frac{x^2+y^2}{w^2}\Big), \qquad C_n(u,v) = \int_{-\infty}^{\infty} \exp\big[i\left(t^n + vt^2 + ut\right)\big]\, dt. \tag{1}$$

Here $n = 4$, the source scale is $s = 0.20$ mm, and the Gaussian envelope radius is $w = 0.30$ mm, as set by the executed optical setup. The source apodization parameter is 0.10. The aperture is circular, with physical radius 0.9 mm.

The quadratic chirp is implemented as

$$V_c(x,y) = V(x,y) \exp\left[ic\frac{x^2+y^2}{w_\phi^2}\right], \tag{2}$$

where $c = +0.5$ and $w_\phi = 0.30$ mm, the actual phase radius from the optical setup. This phase radius is distinct from the Pearcey scale $s = 0.20$ mm. The registered primary input applies the chirp first, then applies radial quartic phase +0.2:

$$U_{\text{in}}(x,y) = V(x,y) \exp\left[i\,0.5\frac{x^2+y^2}{w_\phi^2}\right] \exp\left[i\,0.2\left(\frac{x^2+y^2}{w_\phi^2}\right)^2\right]. \tag{3}$$

The quartic term is a polynomial phase modulation applied to this source, not a canonical catastrophe-beam field. The chirp-only ablation omits the final exponential in Eq. (3).

The sampled input is normalized to power P on a grid with spacing Δx:

$$U_0(x_j, y_k) = \sqrt{\frac{P}{\Delta x^2 \sum_{a,b} |V_{ab}|^2}}\, V(x_j, y_k). \tag{4}$$

In this equation, V denotes the complete field after the specified phase operations and aperture. The normalization gives equal input power to the compared arms; it does not imply equal output peak intensity.

3.2 Propagation and focal-position observable

Scalar propagation uses the angular-spectrum method (ASM):

$$U_z = \mathcal{F}^{-1}\Big[\mathcal{F}(U_0)\, B_z \exp\Big(i2\pi z\sqrt{\lambda^{-2} - f_x^2 - f_y^2}\Big)\Big]. \tag{5}$$

Here λ is the wavelength, f_x, f_y are spatial frequencies, and B_z is the implemented conservative radial bandwidth mask. Its cutoff is

$$\nu_c = \min\left[\lambda^{-1}, \frac{\lambda^{-1}}{\sqrt{1 + (2z/L)^2}}\right], \tag{6}$$

where L is the computational window width. This radial cutoff is not the complete Matsushima passband construction.

The lens arm includes an ideal thin lens of focal length $f = 60$ mm; the free-space arm propagates the identical transformed input without the lens. For $I_z(x_j, y_k) = |U_z(x_j, y_k)|^2$, the primary observable and signed comparison are

$$\begin{aligned} focal_z_m &= \arg\max_{z\in\mathcal{Z}} \max_{j,k} I_z(x_j, y_k), \\ \Delta z_{\text{peak}} &= focal_z_m^{(\text{lens})} - focal_z_m^{(\text{free})}. \end{aligned} \tag{7}$$

The registered interval is $\mathcal{Z} = [0.04, 0.08]$ m, with 11 planes in the primary cases, spaced 4 mm apart. The prediction for the chirp-then-quartic condition is $\Delta z_{\text{peak}} \geq 0$. A maximum selected at an interval boundary does not locate an interior peak and is inconclusive for interpreting a physical focal-position difference.

For completeness, the circular-core energy fraction is

$$C(z) = P^{-1} \int_{x^2+y^2\leq r_c^2} |U_z|^2 \, dA, \qquad \overline{C} = \text{median}_{z\in\mathcal{Z}} \, C(z), \tag{8}$$

where $r_c = 0.11$ mm. This is a separate diagnostic, not the registered primary observable; it is not used to infer the focal-position result.

4 Numerical Methods

The primary calculation uses a 192×192 grid over a 2.4 mm square window, wavelength 633 nm, input power 1 W, optical waist 0.30 mm, focal length 60 mm, and physical aperture radius 0.9 mm. Grid coordinates are

$$x_j = (j - \lfloor N/2 \rfloor)L/N, \qquad y_k = (k - \lfloor N/2 \rfloor)L/N.$$

The canonical integral in Eq. (1) is evaluated using contour rotation $t = q\exp(i\pi/(2n))$, with q sampled uniformly over $[-3, 3]$ and trapezoidal quadrature with 192 nodes. The source-convergence calculation uses 384 nodes.

The two comparison arms use identical transformed inputs, wavelength, aperture, power normalization, grid, and axial planes. The recorded composition parameters are $\theta = 0$, relative phase $\delta = 0$, and amplitude weight $a = 1$; thus the executed comparison is a single input in each arm, not a nontrivial two-field synthesis. In general, coherent composition is normalized as

$$U_c = \sqrt{\frac{P}{\int |U + ae^{i\delta}\mathcal{R}_\theta U|^2 \, dA}} \left(U + ae^{i\delta}\mathcal{R}_\theta U\right). \tag{9}$$

An exactly zero coherent field cannot be normalized, although other noncancelled arms remain measurable. No coherent pairing is used to generate the primary result.

For the primary and chirp-only cases, the supplied confirmation records mark mesh, window, second-window, axial, independent-observable, reference-physics, and coupled-stochastic-realization checks as passed. They also mark quantization, held-out-wavelength, and two noise-probe checks as passed. These are numerical checks, not independent physical replications. The supplied summary does not provide arm-wise discrepancies for an independent propagator, so those particular values cannot be independently reported here.

5 Results and Physical Interpretation

5.1 Registered construction and chirp-only ablation

Table 1 summarizes the sampled focal-position comparisons. For the registered chirp-then-quartic input, the lens-arm maximum is at 0.064 m, and the free-space maximum is at 0.040 m, yielding

Table 1: Sampled axial locations of the brightest transverse pixel in the lens and free-space arms. Differences use the lens-minus-free-space convention. The free-space maxima in the confirmed rows occur at the lower boundary of the evaluated interval.

Input condition	Lens $focal_z_m$ (m)	Free-space $focal_z_m$ (m)	Δz_{peak} (m)
Chirp +0.5, then quartic +0.2	0.064	0.040	+0.024
Chirp +0.5 only	0.060	0.040	+0.020
Quartic +0.2 only	0.060	0.052	+0.008
No added phase	0.056	0.072	−0.016
Chirp +0.9, then quartic +0.7	0.076	0.040	+0.036

$\Delta z_{\text{peak}} = +0.024$ m. The chirp-only ablation gives 0.060 m with the lens and 0.040 m in free space, giving +0.020 m. Both signs agree with the registered nonnegative prediction, and the supplied numerical confirmation checks pass for both conditions.

The free-space maximum is exactly the first sampled plane in both confirmed cases. Under the registered peak-selection rule, these boundary-selected values do not locate interior free-space peaks. Consequently, the positive sampled differences do not establish resolved physical focal-position separations. The chirp-only result is also close to the primary result, so the available observations do not show that the quartic phase contributes beyond the chirp. These separate scalar comparisons do not estimate an interaction.

5.2 Unconfirmed comparisons and classical controls

The quartic-only case gives a sampled difference of +0.008 m, but fails the axial-convergence and robustness checks. The untransformed reference gives −0.016 m and also fails axial convergence and some robustness checks. The stronger chirp-plus-quartic condition gives +0.036 m in the primary record, but its confirmation fails mesh and axial convergence. These are exploratory observations, not validated evidence for a general phase-strength or direction rule. The negative untransformed comparison cautions against extending the positive sign across input conditions.

Optimized classical phase controls provide context, but are not matched comparisons of the registered lens-minus-free-space effect. Under the stated grid, aperture, lens, and power settings, optimized defocus produced an absolute lens-arm $focal_z_m$ of 0.068 m. An optimized quadratic-plus-quartic control produced 0.080 m, at the upper boundary of the observation interval. These controls do not establish superiority or mechanism. They do show that the sampled lens-arm peak positions are not specific to the Pearcey construction.

The principal interpretive limitation is the boundary selection of the free-space peak, rather than the mere fact that a finite sampled maximum was returned. The saved value 0.040 m describes the largest sampled value in the interval, but does not establish whether the corresponding physical maximum lies at that plane or outside the interval. The positive differences are therefore reported as numerical observations, while their physical focal-position interpretation remains inconclusive. Core-energy diagnostics are separate quantities and do not resolve this limitation.

5.3 Figures

Figure 1 shows the saved propagation diagnostic for the primary construction. It is included to display the axial evolution associated with the reported peak selection and the lower-boundary issue. Figure 2 shows the saved summary of recorded comparisons, distinguishing the two confirmed conditions from the unconfirmed exploratory cases. Neither figure establishes a mechanism.

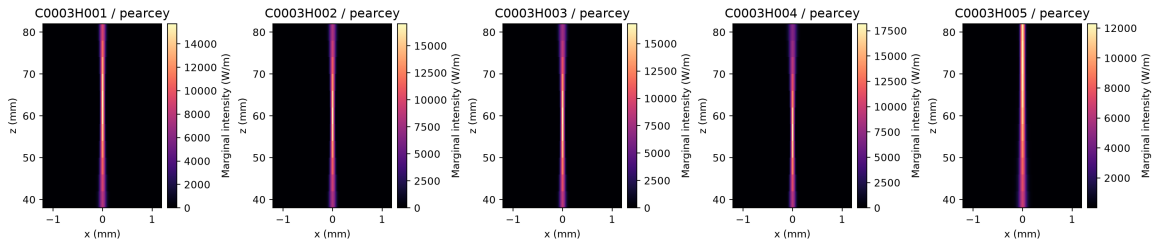


Figure 1: Saved propagation diagnostic for the primary chirp-then-quartic Pearcey condition. The recorded sampled maxima are 0.064 m in the lens arm and 0.040 m in free space. The free-space maximum is at the lower edge of the evaluated interval.

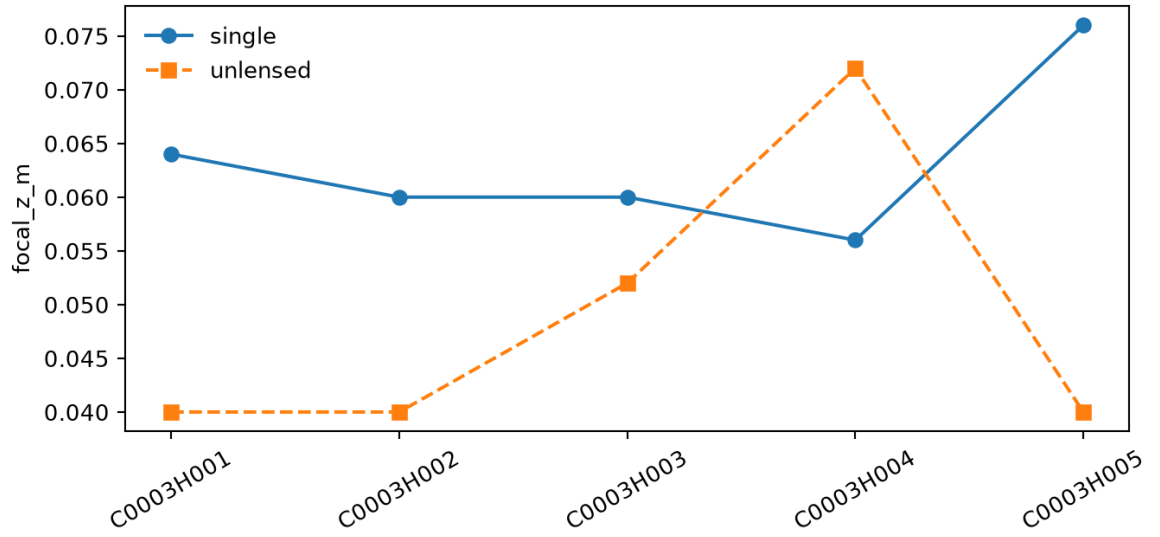


Figure 2: Saved summary of recorded conditions. The primary construction and chirp-only ablation pass the supplied confirmation checks; the quartic-only, untransformed, and stronger phase-control conditions are unconfirmed. The summary does not establish an operator interaction or physical mechanism.

6 Conclusions

For the specified apodized Pearcey source with chirp +0.5 followed by radial quartic phase +0.2, the sampled brightest-pixel locations are 0.064 m with the ideal 60 mm lens and 0.040 m in free space. The resulting +0.024 m difference has the registered sign. The chirp-only ablation also has a positive sampled difference, +0.020 m. Because both free-space maxima are at the lower boundary of the sampled interval, the registered rule leaves the physical focal-position comparison inconclusive. The chirp-only result does not establish an incremental quartic contribution.

These findings apply to the executed sampled-peak endpoint and conditions. They do not demonstrate enhanced concentration, a unique focusing mechanism, or specificity relative to classical phase controls. A discriminating follow-up would first use a predeclared axial range and adequate sampling to resolve interior peak positions in both arms. A matched comparison of no added phase, chirp-only, quartic-only, and the ordered combined phase would then be needed to estimate any interaction.

Acknowledgment

Auto BeamScientist adapts Sakana AI Scientist-v2 [7].

References

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